

RAIN GAUGE

The Tipping Bucket Rain Gauge is a precision instrument designed for accurate and reliable rainfall measurement. It uses a balanced tipping mechanism to record rainfall continuously for meteorological and environmental monitoring applications.

Constructed with durable, weather-resistant materials, it ensures reliable performance in harsh conditions. The gauge supports easy integration with data loggers and monitoring systems for real-time data collection and features a quick-release design for simple maintenance.

Key Features

- Precision Measurement:** Balanced tipping bucket mechanism ensures laboratory-grade accuracy
- Maintenance Optimized:** Minimal moving parts design enhances reliability and reduces servicing requirements
- Advanced Sensing:** Magnetic reed switch technology for precise tip detection
- Intensity Compensation:** Maintains accuracy across varying rainfall rates
- Weather-Resistant Construction:** Durable ABS housing withstands harsh environmental conditions
- System Integration:** Direct compatibility with data loggers and automated weather stations
- User-Centric Design:** Quick-release fasteners for simplified maintenance procedures



Technical Specifications

Parameter	Specification
Instrument Type	Tipping Bucket Rain Gauge
Orifice Diameter	159.5 mm or 200 mm
Collector Area	200 cm ² or 314 cm ²

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Parameter	Specification
Sensing Technology	Magnetic reed switch detection
Sustainability	Up to 5 Year
Response Capability	1 pulse per second maximum operation rate
Measurement Resolution	0.2 mm or 0.5 mm per tip (configurable)
Output Signal	Reed switch pulse count
Measurement Accuracy	±2% (traceable calibration)
Construction Material	UV-stabilized ABS polymer
Range	Up to 150 mm/hour rainfall intensity

Enclosure & Mechanical Specifications

Characteristic	Specification
Body Material	Non-Corrosive material
Protection Rating	IP66 environmental protection
Color	Solar-reflective white
Mounting System	Adjustable base with secure mounting provisions
Maintenance Access	Quick-release fasteners for internal component access
Bucket Material	Injection molded non-hydroscopic ABS (Acrylonitrile Butadiene Styrene)
Levelling	Adjustable leveling screws with integrated spirit level
Debris protection	Integrated stainless steel mesh filter and debris cap system

RAIN GAUGE

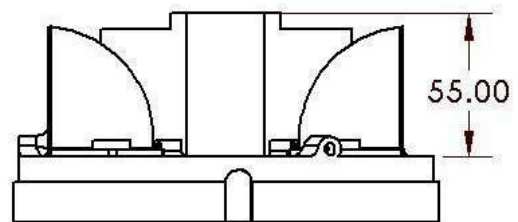
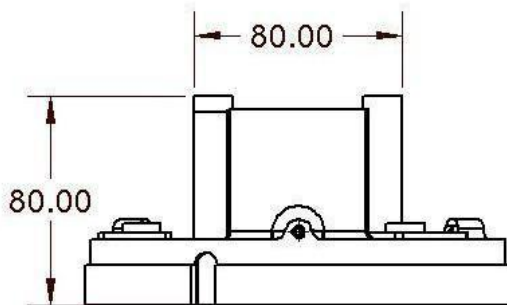
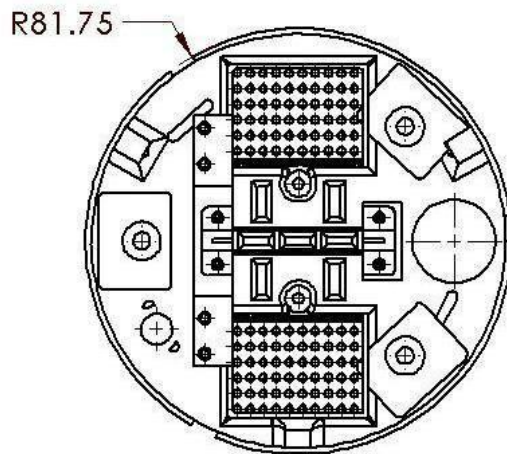
Applications

- **Meteorological Monitoring:** Professional weather station precipitation measurement
- **Agricultural Management:** Irrigation planning and crop water requirement assessment
- **Hydrological Research:** Watershed analysis and rainfall pattern studies
- **Environmental Science:** Climate change research and environmental impact assessment
- **Civil Engineering:** Urban drainage design and flood prediction systems
- **Educational Institutions:** Atmospheric science research and teaching applications

Quality Assurance

- Factory calibrated with traceable certification
- Compliance with international meteorological standards
- Environmental testing for temperature and humidity extremes
- Vibration and shock resistance validation
- Long-term durability testing for UV and weather resistance

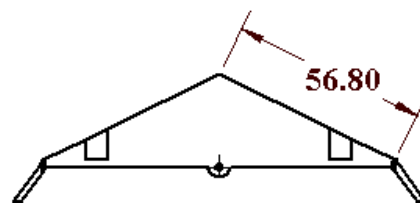
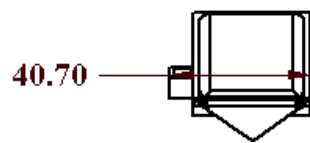
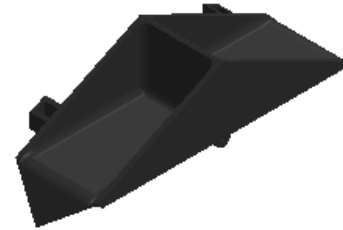
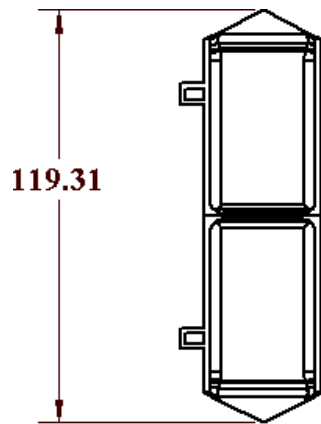
RAIN GAUGE



Rain Gauge Base

(All The Dimensions are in mm)

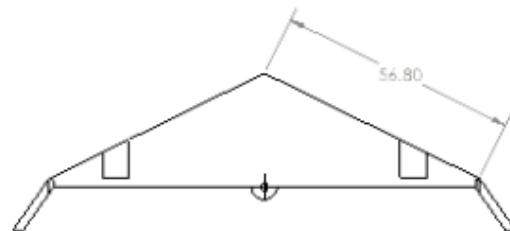
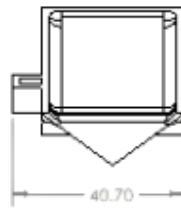
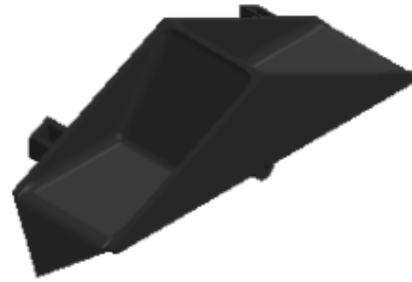
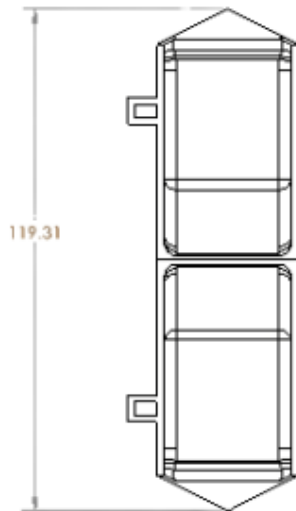
RAIN GAUGE



Rain Gauge SeeSaw (for resolution 0.5)

(All The Dimensions are in mm)

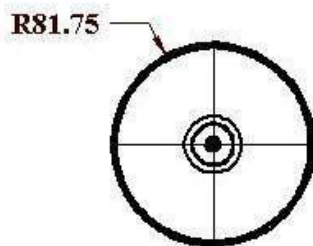
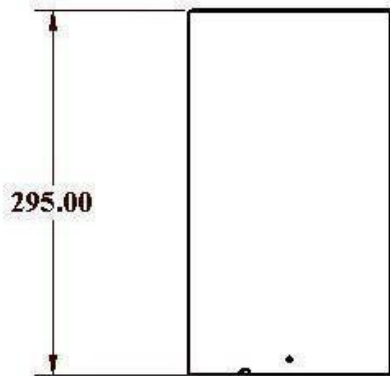
RAIN GAUGE



Rain Gauge SeeSaw (for resolution 0.2)

(All The Dimensions are in mm)

RAIN GAUGE



Rain Gauge Cylinder

(All The Dimensions are in mm)